A Two-Phase Curriculum for Goal-Conditioned Visitation Values from Manipulation Play Data

1 Setting

Offline data $\mathcal{D}$ of unlabeled "play" trajectories from OGBench `cube-single-play-v0` (Park et al., 2024) (state observations; a robot arm and one cube; ∼1M transitions). The goal of the program is zero-shot RL in the sense of Touati and Ollivier (2021); Touati et al. (2023): train once on $\mathcal{D}$ without rewards; at test time, given a reward function (specified by labeled samples), produce a policy with no further training. The central object is the successor measure

$$M^\pi(s,a,X) \;=\; \mathbb{E}_\pi\Big[\textstyle\sum_{t\ge 0}\gamma^t\,\mathbf{1}[s_{t+1}\in X]\;\Big|\; s_0=s,\,a_0=a\Big],$$

the discounted occupancy of state sets X (Dayan, 1993; Blier et al., 2021). Given M^π for the right policies, any reward r yields $Q_r(s,a)=\int r\,dM^\pi(s,a,\cdot)$, hence zero-shot evaluation and (with a policy-improvement step) zero-shot control. FB representations (Touati and Ollivier, 2021) learn a factorization

$$M^{\pi_z}(s,a,X)\approx\int_X F(s,a,z)^\top B(x)\,dx,$$

in which the *policy index* z and the *queried future state* x are separate arguments: one fixed continuation policy π_z, occupancy of every x under it.

Scope of this note. What we train below is not yet this full object. It is the *diagonal slice* $g\mapsto M^{\pi_g}(s,a,B_\epsilon(g))$ — the visitation value of a goal region under that goal's own policy (Section 9). This suffices for goal-reaching tasks and for composition over small candidate-goal sets, and it is where the training-stability content of the note lives; extending it to the separated (z,x) form needed for arbitrary rewards is ongoing work (Section 10).

The problem. On this domain, methods built on future-state reachability — FB, successor features, contrastive value functions — sit far below goal-conditioned TD baselines trained on the same data with the same actor class: FB variants plateau around 0.42 overall success while goal-conditioned IQL (Kostrikov et al., 2021) with a bilinear critic reaches 0.90. The gap is not exploration or data: everything is offline and shared. Something about the successor-measure *objective* fails here.

2 What the goal-conditioned baseline gets right

The strongest method on this data is not a successor-measure method at all: goal-conditioned IQL with a *bilinear* critic $Q(s,a,g)=\phi(s,a)^\top\psi(g)/\sqrt{d}$, trained by expectile TD on the sparse $0/-1$ goal-reaching reward with hindsight-relabeled goals, reaches 0.904. This baseline settles two questions before we change anything:

1. **The factored representation is not the bottleneck.** The bilinear critic *is* the rank-d factorization that successor-measure methods use: $\psi(g)$ is a goal embedding playing the role of B, $\phi(s,a)$ plays the role of F. It trains to 0.904 with $d=50$ on this data. This rules out capacity explanations *for the goal-reaching slice* of the problem: a rank-50 bilinear form demonstrably suffices for it. It

does not rule out capacity limits for the full successor measure — the full object can require much higher rank than any one slice — but it shifts the burden: whatever fails when the same architecture is trained as a measure, fails at least in part in the *training signal*.

2. **The grounding discipline.** Every ingredient of the baseline's objective is anchored: the target reward is a representable, bounded 0/1 event (arrival at the goal); positives come hindsight-paired with their own trajectory future; the reward term pins the scale of the bilinear product, so no auxiliary regularizer on ψ is needed. None of these properties hold for the density-based successor-measure TD (Section 3).

Phase 1 of our scheme can be read as importing exactly this discipline into the measure setting. Indeed the two objects are affinely related: under the absorbing $0/-1$ convention the optimal value satisfies $V = -\big(1 - \mathbb{E}[\gamma^T]\big)/(1-\gamma)$, i.e. $\mathbb{E}[\gamma^T] = 1 + (1-\gamma)V$, so the reaching value of phase 1 is the baseline's value function re-expressed in measure units ($[0,1]$, probability-like), with the $n \times n$ matrix supervision of Section 6 as the only structural addition. What the baseline does *not* provide is occupancy semantics: its value saturates on arrival (no revisit mass), so integrating a general reward against it, $\int r\, dM$, is not meaningful. That missing piece — and only that piece — is what phase 2 adds.

3 Intuition: occupancy factors into reaching × recurrence

Write $B_\epsilon(g)$ for a small ball around a goal state g (we use the cube position, $\epsilon = 4\,\text{cm}$). Let $T_g = \min\{t \geq 1 : s_t \in B_\epsilon(g)\}$ be the (random) time the region is first reached. The discounted occupancy of the ball splits at T_g:

$$M(s, a, B_\epsilon(g)) \;=\; \underbrace{\mathbb{E}\big[\gamma^{T_g}\big]}_{Q(s,a,g)} \;\cdot\; \underbrace{\mathbb{E}\big[\text{discounted occupancy of } B_\epsilon(g) \text{ accumulated from arrival}\big]}_{R(g)}, \qquad (1)$$

Figure 1 sketches the decomposition. Two qualifications are needed before using it. First, the split is *exact* only when the post-arrival occupancy is the same for every way of entering the region — true for a singleton state of a Markov chain, only approximate for a region. Second, and specific to manipulation: our region is defined on the *cube position only*, so two arrivals at the same cube position can differ in grasp state, arm configuration, and cube velocity — a secure placement and an unstable pass-through have the same arrival time but very different subsequent occupancy. The action-independence of the second factor is therefore an *empirical hypothesis* about this data, not a consequence of the Markov property, and the recurrence term may carry real control information (stable vs. transient arrivals) that the decomposition deliberately sets aside. Useful diagnostics we have not yet run: the conditional variance of M/Q given g, and the agreement rate of $\arg\max_a Q$ with $\arg\max_a M$.

Two empirical facts about play data make this decomposition load-bearing:

1. **The recurrence factor is large and nearly action-independent.** Play trajectories dwell and revisit: the Monte-Carlo estimate of the second factor on our data has mean ≈ 24.7 (in units of discounted visits, $\gamma = 0.99$) with large variance across goals. Fitting a goal-only regressor $R(g)$ explains occupancy well (rank correlation ≈ 0.6 with Monte-Carlo occupancy, uniformly across horizon buckets), while the reaching term alone explains almost none of it (≈ 0.06). So most of the *magnitude* of occupancy carries no control signal.

2. **The standard estimator spends its capacity on exactly that part.** The density-based TD used by FB-style methods handles the on-diagonal (observed successor) term *linearly* — the loss contains $-\mathbb{E}[M]$, a term that always pays for making M larger — and is stabilized only by an orthonormality regularizer on the state embedding. On this domain that regularizer is itself destructive (below). Removing it un-anchors the linear term: we verified that even with a *frozen, known-good* goal embedding, the density-based TD inflates without bound (critic scale $154 \to 5\times10^4$ within 65k steps) and control degrades monotonically ($0.98 \to 0.42$ by 100k steps).

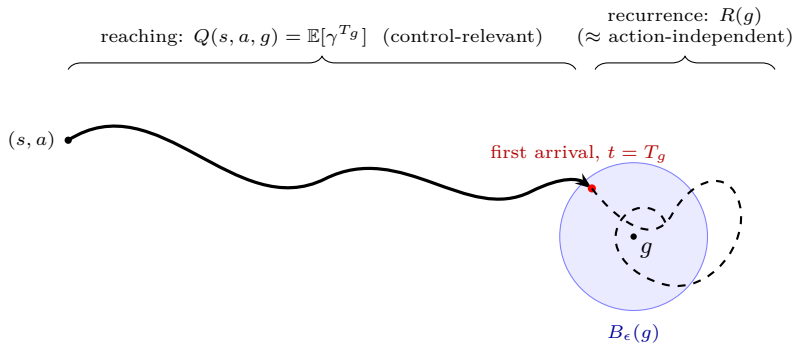


Figure 1: The occupancy of a small goal region splits at the first arrival: everything before T_g is the *reaching* problem (solid), which is what control must solve; everything after (dashed: dwelling and revisiting, prominent in play data) contributes only the goal-dependent recurrence mass $R(g)$. Eq. 1: $M = Q \cdot R$. On our data R has mean ≈ 24.7 discounted visits — most of the magnitude of M carries no control signal.

The design that falls out: *learn the reaching term first, with a bounded regression objective; then learn occupancy on top of it, using an objective in which every term is a squared residual against a representable* $0/1$ *reward.*

4 From density-ratio TD to finite-region regression

Section 3 argued *empirically* that the density estimator spends its capacity on recurrence and inflates without an external anchor. This section derives *why* from the successor-measure Bellman equation, and shows that a single substitution — a finite goal region in place of a point query — turns every entry of the training matrix into a bounded squared residual, which is what phases 1 and 2 exploit. It also states the objects in the separated $F^\top B$ form, connecting the diagonal slice trained below (Section 6) to the full FB object (Section 10).

The singular immediate term. Write $m_z(s,a,x) = dM^{\pi_z}(s,a,\cdot)/d\rho\,(x)$ for the density of the successor measure relative to a reference ρ; this is the object FB factorizes as $F(s,a,z)^\top B(x)$. It satisfies the density Bellman equation

$$m_z(s,a,x) \;=\; \frac{\delta_{s'}(x)}{\rho(x)} \;+\; \gamma\, \mathbb{E}_{a' \sim \pi_z(\cdot|s')}\big[m_z(s',a',x)\big].$$

The immediate term is a *Dirac density ratio*: on the observed successor it is a point mass divided by ρ, not a finite scalar that can be placed inside a squared Bellman residual. FB therefore does not regress this equation directly. It minimizes the least-squares density-matching surrogate

$$\tfrac{1}{2}\,\mathbb{E}_{x\sim\rho}\big[(m_z - \gamma m_z')^2\big] \;-\; \mathbb{E}\big[m_z(s,a,s') - \gamma m_z'(s,a,s')\big] \;+\; \lambda_{\text{ortho}}\,\mathcal{L}_{\text{ortho}},$$

in which the off-diagonal (random-query) entries carry the quadratic residual and the observed successor enters only through the *negative linear* term — the same $-\mathbb{E}[M]$ flagged in Section 3. This term has no fixed target; it pays unconditionally for making the matched-successor value larger, and its scale is held in check *only* by the orthonormality regularizer on B. Remove that regularizer and nothing bounds the diagonal — precisely the divergence we observe on a frozen, known-good embedding (critic scale $154 \to 5 \times 10^4$; Section 7.5). The two standard stabilizers are thus load-bearing in opposite directions: ortho anchors the scale but destroys the reaching geometry (the 2×2 of Section 7.4).

The substitution. Replace the point query by a finite region. With the cube-position ball $B_\epsilon(g)$ and the hit indicator $h(s', g) = \mathbf{1}[\|\text{cube}(s') - \text{cube}(g)\| \le \epsilon] \in \{0, 1\}$ of Section 6, the singular immediate term $\delta_{s'}(x)/\rho(x)$ becomes the bounded scalar $h(s', g)$. Every entry of the matrix — diagonal included — now has an explicit target and is an ordinary squared residual: no separate linear diagonal term, no off-diagonal mask, and the 0/1 event pins the scale directly, so no orthonormality (or any) regularizer on B is needed. This substitution is the whole content of the change,

$$\text{density-ratio estimation} \longrightarrow \text{Bellman regression on finite goal regions.}$$

The two phases are then the same regression under two backups, differing only in whether arrival absorbs:

$$y_{ij}^{\text{reach}} = h_{ij} + \gamma(1 - h_{ij})\,\bar{Q}_{ij}, \qquad y_{ij}^{\text{visit}} = h_{ij} + \gamma\,\bar{M}_{ij}.$$

Phase 1 (Algorithm 1) terminates the backup at first arrival, giving the bounded reaching value $Q = \mathbb{E}[\gamma^{T_g - 1}] \in [0, 1]$; phase 2 (Algorithm 2) drops the $(1 - h_{ij})$ factor so revisits keep accumulating, giving the visitation value $M \in [0, 1/(1 - \gamma)]$. The kernel event is what makes both objects bounded and self-anchored — the property the density form structurally lacks.

Same architecture, separated arguments. Nothing above ties the policy index to the query. In full FB form the two objects are $Q(s, a, z, x)$ and $M(s, a, z, x) = F(s, a, z)^\top B(x)/\sqrt{d}$ with z and x *independent*; the diagonal slice $z = \psi(g)$, $x = g$ trained in Sections 6–7.7 is the special case in which the policy and the query name the same goal. The reach-first substitution therefore applies verbatim to the separated form of Section 10 — only the batch's row-index (z_j) and column-query (x_k) sampling changes, not the loss.

5 The structure in a 50-state playpen

Before the algorithm, the structural claims of Sections 3 and 4 can be exhibited *exactly* in a minimal MDP built to have the same anatomy as manipulation play data (`scripts/rescuing_fb/tabular_playpen.py`; everything below is a linear solve unless marked SGD).

The playpen MDP. A 7×7 grid plus a pit (50 states, 5 actions: stay/up/down/left/right). The *semantic coordinate* is the **column** (the cube-position analog: goal regions are columns); the *interaction mode* is the **row** (the grasp/velocity analog, invisible to the region). A 2×2 **bowl** is sticky — move attempts leave with probability 0.1 (a cube at rest); the top row is a **conveyor** that drifts two columns per step regardless of the action (a carried cube: fast transport, but position cannot be held), and riding off its end drops into a hard-to-leave pit (a dropped cube must be re-fetched). The "play" behavior policy is a goal-directed mixture of reach-a-column policies plus exploration noise; $\gamma = 0.99$.

(i) The factorization is exact, and recurrence carries the magnitude. For singleton goals the renewal identity $M = Q \cdot R$ of Eq. 1 holds to numerical precision (max error 1.4×10^{-9}). Recurrence spans a wide range ($R \approx 16.5$ in the bowl vs. 1.9 on the conveyor) while reaching values compress toward 1 at this horizon. Across the seven column regions under the play policy, a *goal-only* recurrence predictor ranks occupancy at Spearman **0.96** while the reaching term alone ranks it at 0.58 (R spans 4.2–30.7; reaching spans 0.34–1) — the playpen version of the 0.60-vs-0.06 statistic of Section 7.5, here exact rather than estimated.

(ii) Arrival ranking and occupancy ranking genuinely disagree. Weight two regions — the bowl (stable) and one conveyor cell (transient) — by w_{bowl}=1, w_{conv}=2.11, chosen between the reaching ratio (1.04: the conveyor cell is nearly as *reachable*) and the occupancy ratio (4.26). Ranking candidates by weighted reaching value picks the conveyor; ranking by weighted occupancy picks the bowl; exact policy evaluation confirms the bowl (96.8 vs. 47.9 discounted return). A goal-reaching value, however well trained, selects the wrong region here by construction — this is the tier-3 task of the evaluation suite (spec 11) in miniature. The diagonal-slice defect of Section 9 is also exact: summing the two slices values the pair at 144.7, while no single policy can achieve more than 96.8.

(iii) The bounded regression dominates the density surrogate at matched rank (SGD). Fitting rank-8 bilinear $F^\top B$ on 20k offline play transitions (fixed policy, $\gamma = 0.95$, Adam, identical

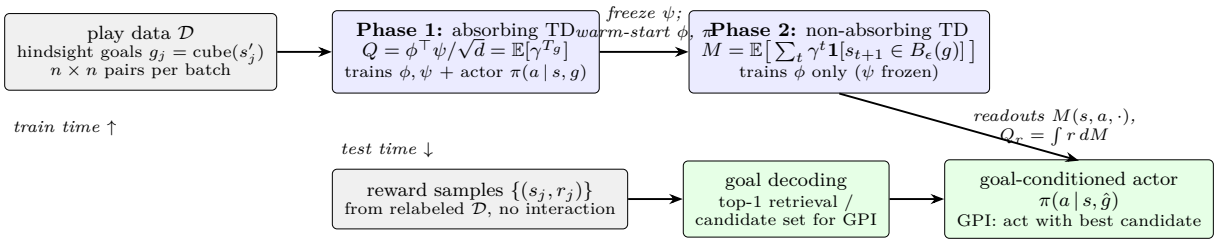


Figure 2: Pipeline. Training (top): phase 1 learns the goal-reaching value and the policy; phase 2 re-trains the forward tower ϕ into a visitation measure on the frozen goal embedding ψ. Test time (bottom): a reward specified by labeled dataset samples is reduced to one or a few candidate goals (retrieval / GPI), executed by the goal-conditioned actor; the measure readouts score candidates and integrate rewards.

architecture and data): the column-region visitation regression reaches relative error 0.05–0.09 and per-state goal-ranking correlation 0.96–0.99; the FB density-TD reaches only 0.38 / 0.65, and *adding* the orthonormality regularizer makes it worse (0.53 / 0.45) — ortho damaging the geometry exactly as in the 2×2 of Section 7.4. Two honest notes: with no actor in the loop the density estimator here *fits poorly but does not diverge* (its output scale settles $\sim 8\times$ the region form's, with a target dynamic range of ~ 80 vs. a bounded target of ≤ 12), consistent with the divergence of Section 7.5 being a property of the estimator-plus-actor system; and absorbing-first curriculum and direct visitation regression perform alike in this pure-evaluation setting (0.09 vs. 0.05), consistent with the curriculum's value lying in the control loop, not in policy evaluation.

6 Algorithm

Figure 2 shows the full pipeline; the two phases are specified in Algorithms 1 and 2.

Both phases use the bilinear critic

$$Q_\theta(s, a, g) = \phi(s, a)^\top \psi(g) / \sqrt{d}$$

(two-layer MLP, $d = 50$, twin heads with min-backup), and both are trained on minibatch "matrices": for a batch of n transitions (s_i, a_i, s'_i), the columns are the n relabeled goals $g_j = \text{cube}(s'_j)$ (hindsight relabeling (Andrychowicz et al., 2017)), giving $n \times n$ residuals per batch — the diagonal pairs each transition with its own future, the off-diagonal supplies random goals. The policy is a flow-matching behavior-cloning policy with one-step distillation and a greedy term on the critic readout, following flow Q-learning (Park et al., 2025); the greedy term is normalized by the (stop-gradient) magnitude of Q to be scale invariant.

Algorithm 1 Phase 1 — goal-reaching value (absorbing formulation)

for each gradient step **do**
  sample batch $\{(s_i, a_i, s'_i)\}_{i=1}^n$; goals $g_j = \text{cube}(s'_j)$; $h_{ij} = \mathbf{1}[\,\|\text{cube}(s'_i) - g_j\| \leq \epsilon\,]$
  $a'_i \sim \pi(\cdot \mid s'_i, g)$ ▷ bootstrap action from the current policy
  $y_{ij} = h_{ij} + \gamma\,(1 - h_{ij})\,\bar{Q}(s'_i, a'_i, g_j)$ ▷ absorbing: reaching the ball ends the backup
  $\mathcal{L}_Q = \frac{1}{n^2} \sum_{ij} \big(Q(s_i, a_i, g_j) - \text{sg}[y_{ij}]\big)^2 \,/\, \text{sg}[\overline{Q^2}]$
  update ϕ, ψ on $\mathcal{L}_Q$; update policy (flow BC + distilled greedy step on Q)
end for

The fixed point is $Q(s, a, g) = \mathbb{E}[\gamma^{T_g - 1}] \in [0, 1]$ (the immediate-hit case T_g=1 receives target 1, so the discount counts from the current transition; this is the convention under which the affine identity of Section 2, $Q = 1 + (1 - \gamma)V$, is exact). The sparse 0/1 arrival reward is representable, so *every* entry

— diagonal included — is a plain squared residual; the idealized backup operator is a γ-contraction; the reward pins the scale, so no orthonormality (or any other) regularizer on ψ is used.

Algorithm 2 Phase 2 — visitation measure on the frozen goal embedding

initialize all networks from phase 1; **freeze** ψ
for each gradient step **do**
  sample batch, goals, h_{ij} as in phase 1
  $y_{ij} = h_{ij} + \gamma \bar{M}(s'_i, a'_i, g_j)$ ▷ *non*-absorbing: revisits keep counting
  $\mathcal{L}_M = \frac{1}{n^2} \sum_{ij} \left(M(s_i, a_i, g_j) - \mathrm{sg}[y_{ij}]\right)^2 / \mathrm{sg}[\overline{M^2}]$
  update ϕ (and the policy; greedy term normalized by the *centered* magnitude $\mathrm{sg}\,\mathbb{E}|M - \bar{M}|$)
end for

The phase-2 fixed point is the discounted visitation count

$$M(s, a, g) = \mathbb{E}\left[\sum_{t \geq 0} \gamma^t \mathbf{1}[s_{t+1} \in B_\epsilon(g)]\right]$$

— a visitation value (occupancy of the ball, revisits counted), bounded by $1/(1-\gamma)$, with squared residuals only. The contraction statement applies to the idealized fixed-policy evaluation operator; the practical estimator (changing actor, rank-50 bilinear approximation, twin-min backups, offline bootstrap actions, a ψ trained for a different target) carries no such guarantee — stability is an empirical finding here, not a theorem. Freezing ψ carries the phase-1 goal geometry into phase 2 and isolates what changed if phase 2 misbehaves. The centered actor normalization compensates for the fact that, per Eq. 1, M carries a large shared recurrence baseline that would otherwise dilute the greedy signal. (A variant we have not yet run keeps the phase-1 *actor* frozen too and trains only a separate visitation head: if the recurrence factor really is action-independent, phase-2 actor updates can only add drift, and this variant would isolate that.)

An alternative phase 2, which requires no TD at all: keep Q from phase 1 and fit the scalar recurrence factor $R(g)$ of Eq. 1 by Monte-Carlo regression on the frozen $\psi(g)$, using $M \approx Q \cdot R$. This is cheaper but only approximate; the TD version is exact in the limit and is our primary variant.

7 Experiments

7.1 Setup, protocol, and nomenclature

Environment and data. All experiments use OGBench `cube-single-play-v0`: a 6-DoF arm and one cube, 28-dimensional state observations, and an offline dataset of 10^6 transitions collected by a scripted "play" policy that repeatedly picks the cube up, moves it, and sets it down (Figure 1 and the report strips show what this looks like). No reward labels are used at training time.

Training. Unless stated otherwise: bilinear critic $\phi(s, a)^\top \psi(g)/\sqrt{d}$ with d=50, twin heads with min-backup, $\gamma = 0.99$, goal ball $\epsilon = 4\,\mathrm{cm}$ on the cube position, hindsight-relabeled batch goals ($n \times n$ residuals per batch), flow-BC + one-step-distillation actor, 500k gradient steps, 2–3 seeds per cell. Curves are evaluated every 50k steps.

Evaluation. Five harnesses are used in this note, measuring different things; Table 1 is the index, and the two that carry the headline numbers are described immediately below.

(i) *Native goal-conditioned evaluation*: the five OGBench `cube-single` tasks, 50 episodes per task, overall success = mean task success; this is the number reported for every training curve and final checkpoint. (ii) *Zero-shot harness*: a task is specified only by reward labels on m=2000 states relabeled from the offline dataset; the harness infers a task representation, extracts a policy without any training, and rolls out 25 episodes per task variant at horizon 200 in batched lock-step environments. The six reward families are defined in Table 2 and visualized on the cube workspace in Figure 3; $c(x)$ denotes the cube position of state x, c^* a task-goal cube position, and $\epsilon = 4\,\mathrm{cm}$ throughout.

Harness	What it scores	Protocol	Where
Native GC	the *actor*, on the 5 built-in tasks	50 episodes/task, mean success	everywhere
Zero-shot (goal-decode)	inference + decode over 6 goal-reducible families	m=2000 labels, 25 ep., H=200	§7.12, Tab. 2
Set-mass calibration (L1)	whether $\tilde{M}$ *is a measure*	32 starts × 3 rollouts vs. MC, 3 box scales	§7.8
Novel-reward transfer (L2)	the *readout*, on 48 unseen rewards	prediction + control, 3 baselines	§7.9, Tab. 3
Measure benchmark (SMB)	measure quality across variants	Wing-B: B1–B3	§11

Table 1: The evaluation suite. The first two score behavior, the last three score the object. They are not interchangeable, and §7.8 shows they can disagree.

Family	Reward $r(x)$	Success criterion	What we see
DIRAC	$\mathbf{1}[\|c(x) - c^*\| \le \epsilon]$	environment success flag at c^* (identical to the native eval)	solved: 0.89–0.97 retrieval, 0.90/0.92 direct-z
MIXTURE	$\max_{k\in\{1,2\}} \mathbf{1}[\|c(x) - c_k\| \le \epsilon]$	success at *either* goal (all 10 task pairs)	0.64–0.89 family mean; easiest pairs 0.94–1.00
REGION	$\mathbf{1}[c(x) \in R]$, R = 6 cm box or 4 cm ball	cube inside R at episode end	0.73–0.80 retrieval — below every goal-reducible family
DENSE	$\exp\big(-\|c(x) - c^*\|/\sigma\big)$, σ=8 cm	environment success flag at the peak c^*	0.93–0.99 retrieval; direct-z seed-split 0.90/0.13
LINEAR	$\psi(x)^\top \tilde{z}/\sqrt{d}$, random unit $\tilde{z}$	no success flag exists; achieved discounted reward vs. a random-z control	unresolved: beats control on 2/3 phase-1 seeds, enormous variance, no edge on the separated form
REACH-AVOID	$\mathbf{1}[c(x) \in A] - \mathbf{1}[c(x) \in B]$	success-at-A; fraction of episode time in B (exploratory)	reach side 0.91–0.96 retrieval; avoid side non-discriminative (time-in-B $\le$ 0.003 for *every* policy)

Table 2: The zero-shot reward families: definition, scoring, and the one-line observation. Per-route, per-seed numbers in Table 7.12.

Inference variants: *moment/regression* z_r in embedding space (consumed directly or decoded); *retrieval-k* (mean embedding of the top-k reward-weighted dataset states; k=1 decodes to the single best state as a goal); *GPI* over a candidate set (act with the candidate whose critic readout is highest in the current state).

All numbers below are overall success on this protocol unless noted.

7.2 Anchors

Goal-conditioned IQL with the same bilinear critic and the same actor class reaches **0.904**; the FB representation trained on the same data with the same actor class plateaus at ≈ 0.42. These two numbers frame everything below: the gap is the phenomenon, and 0.90 is the bar a measure-trained critic must meet.

7.3 The measure-objective ladder (the negative record)

Before proposing anything, we ran the canonical repairs, each as a single-axis move on the same measure-only cell (bilinear d=50, state-action measure on the critic legs, 3 seeds each unless noted). Table 7.3 is

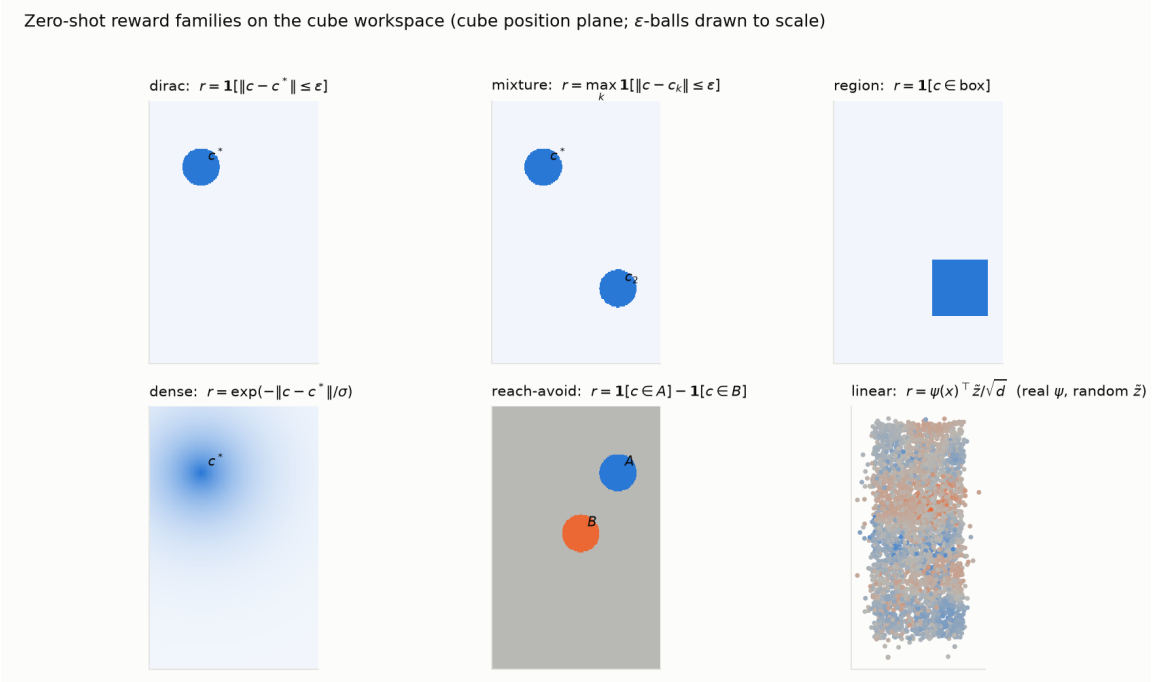


Figure 3: What each reward family looks like on the cube workspace (cube-position plane; ϵ-balls to scale). The first five are functions of cube position. The LINEAR panel is computed with the *trained* phase-1 embedding: dataset states colored by $\psi(x)^\top \tilde{z}$ for one random direction $\tilde{z}$ — the field is spatially incoherent in cube space, which is the visual form of the extraction boundary: no reduction of this reward to a goal (or any small candidate set) exists, so only genuine measure integration can serve it.

the complete record.

The single informative rung: changing the *semantics* of the target from occupancy to first arrival (0.225, sustained, despite a $\sim$100$\times$ oversized regularizer in the changed units). That rung became phase 1.

7.4 Phase 1: the reaching value — the deciding 2×2

The two candidate explanations for 0.225 vs. the 0.90 bar were the target semantics and the orthonormality regularizer. The 2×2 (3 seeds per cell, 500k):

	no regularizer	with regularizer
reaching semantics	$\mathbf{0.967 \pm 0.012}$ (0.956/0.964/0.980)	0.225
occupancy semantics	floored / unstable	≈ 0.10

Both ingredients are necessary; neither suffices. The winning cell crosses 0.94 by 50k steps and is stable to 500k (Figure 4). **Verification.** Because a single surprising cell can be an evaluation artifact, each of the three checkpoints was re-evaluated in a separate process with fresh environment seeds (100 episodes/task for seed 0, 50 for seeds 1–2): 0.972/0.988/0.984. Six additional incidental step-0 re-evaluations during later warm-starts read 0.96–0.99.

7.5 Mechanism evaluations

The frozen-ψ divergence experiment. Warm-start every network from a finished phase-1 checkpoint, freeze ψ, and train ϕ with the density-form TD (the FB estimator: linear observed-successor term, squared

Variant	Success	Note
density-TD occupancy (FB form, with orthonormality)	≤ 0.10	the starting point
+ conservatism (min-aggregation)	0.240	pessimism *hurts*
faithful one-step variant	0.067	$\approx$ BC floor
hindsight test-point columns alone	0.095	
in-sample expectile backup alone	0.103	
paired↔random column dial, deconfounded[a]	0.00	every setting flat
reaching semantics alone (regularizer still on)	0.225	the first cell off the floor
task-embedding grounding mixed into the critic	collapses	0.96 mid-run $\rightarrow$ 0.0–0.4 (7/7 seeds)
embedding-conditioned actor under a sampling prior	0.395	the prior, not the conditioning (§7.13)

[a] rerun with the regularizer off and the loss-scale defect fixed, so the earlier confound is removed; includes the full-corner combination (reaching + expectile + hindsight columns), which also sits at 0.00 — stacking axes on the reaching cell *destroys* it.

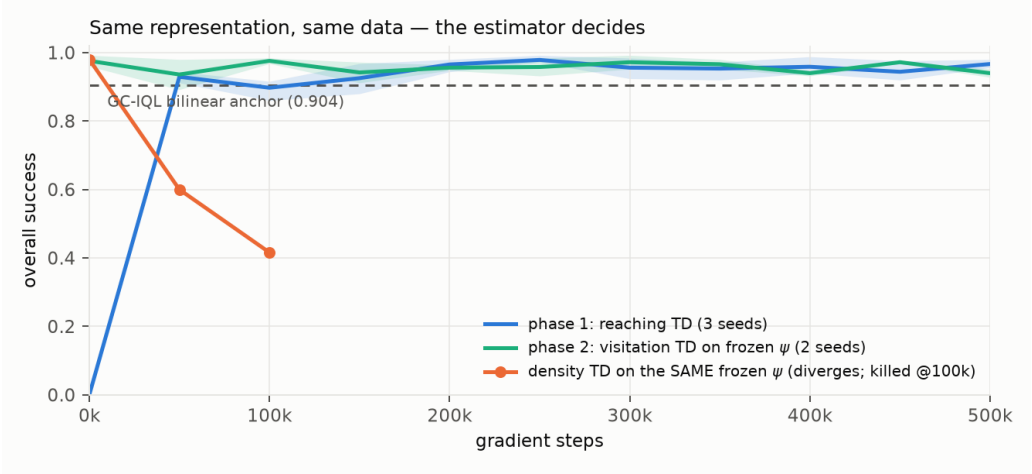


Figure 4: Native-evaluation success against gradient steps. One representation, one dataset, three training signals: the reaching TD (blue, 3 seeds) and the visitation TD on the frozen reaching geometry (aqua, 2 seeds) sit at the GC-IQL anchor for the full run; the density-form TD on the *same frozen geometry* (orange) degrades immediately and monotonically.

off-diagonal term, no regularizer). If the historical failures were representational, this cell must succeed — the representation is, by construction, one that supports 0.97 control. Instead the critic scale inflates $154 \rightarrow 5.4 \times 10^4$ within 65k steps and success decays $0.98 \rightarrow 0.60 \rightarrow 0.42$ by 100k (3 seeds; Figure 5). A control with ψ *trainable* collapses faster still ($0.97 \rightarrow 0.07$ and $0.98 \rightarrow 0.06$ at 50k, 2 seeds): representation corruption compounds the estimator defect but is not its cause.

Revisit-mass statistics. Monte-Carlo, over the dataset at the 4 cm goal ball: discounted occupancy accumulated *after* first arrival has mean ≈ 24.7 (within-trajectory estimate ≈ 30, bimodal: "passed through" vs. "settled"), with large variance across goals. A goal-only regressor $R(g)$ of this mass, fit on the frozen $\psi(g)$, predicts Monte-Carlo occupancy at Spearman rank correlation 0.60–0.63 *uniformly across horizon buckets* (including first-arrival times > 20 steps), while the reaching term alone predicts at 0.06; a position-only regressor fits 13$\times$ worse (MSE 0.93 vs. 0.069) than the ψ-input one. This is the quantitative content of Eq. 1: the recurrence factor carries essentially all of occupancy's magnitude, and the learned embedding supports predicting it.

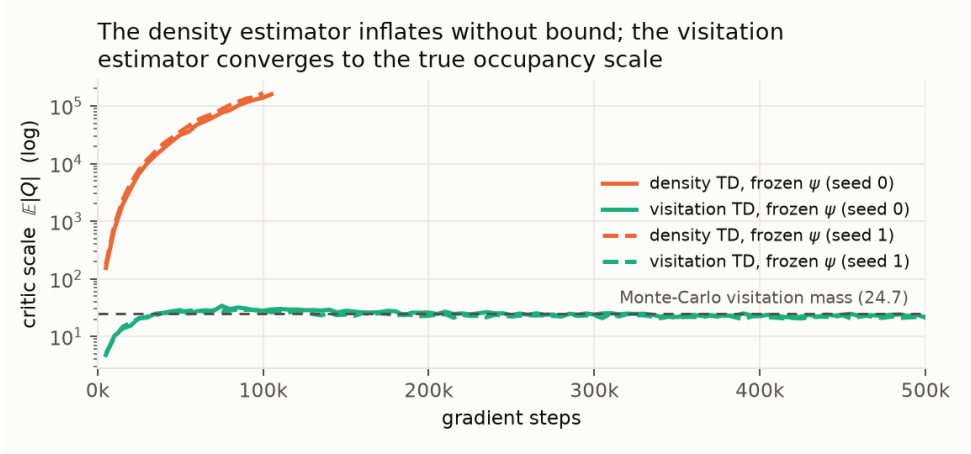


Figure 5: Critic scale $\mathbb{E}|Q|$ (log) during phase-2-style training on the frozen phase-1 embedding. The density estimator (orange) inflates without bound — its linear term pays unconditionally for larger M once the orthonormality anchor is removed. The visitation estimator (aqua) plateaus on the Monte-Carlo estimate of the true discounted visitation mass (dashed): it converges to the correct measure, not a rescaled surrogate.

7.6 Replication in the FB codebase, and what the goal region is doing

Every result above was produced in one implementation. We re-implemented the phase-1/phase-2 objective a second time, inside the FB reference codebase (PyTorch) rather than the goal-conditioned one (JAX), inheriting FB's own flow actor, latent sampling, target networks and soft updates unchanged; the sole edit is the successor residual. This also switches the evaluation protocol: the numbers in this subsection are FB's *zero-shot* harness, where z is inferred from 10^4 reward-relabelled offline transitions, not the native goal-conditioned eval used elsewhere in this note. Cross-framework numbers are therefore **not** interchangeable with the 0.967 of Section 7.4; within this subsection all rows share one protocol and one anchor, and only those comparisons are made.

Re-implementation also lets us close an ablation this note previously lacked. The working cell differs from stock FB along *two* axes at once, and the caveat in Section 12 concedes the two were never separated:

(A) **reward support.** FB places a Dirac on the exact next state — the identity diagonal of the $[B\times B]$ goal matrix. We replace it with the ϵ-ball $h(s', g) = \mathbf{1}[\|c(s') - c(g)\| \le \epsilon]$, whose radius (4 cm) matches the evaluation reward's own threshold.

(B) **loss form.** FB's quadratic-offdiagonal + *linear*-diagonal split, with orthonormality and an unbounded fixed point, becomes an every-entry-squared bounded residual with no regularizer.

Only (B) is a novel claim, so (A) needs its own control: *FB's exact loss, with the Dirac support generalized from I to the ball, everything else stock* — linear diagonal, orthonormality on, fixed point unbounded. With $h = I$ this control reduces algebraically to the stock FB loss (the off-diagonal mask becomes $1 - I$ and the linear term $-\sum_{ij}(\mathrm{diff}\odot I)/\sum I$ recovers FB's $-\overline{\mathrm{diag}}\cdot n_{\mathrm{par}}$), so it is a strict one-axis generalization rather than a re-implementation; that identity is asserted as a unit test.

arm	support	loss	mean success	per seed
stock FB (anchor, 10 seeds)	I	FB	0.420 ± 0.055	—
control (3 seeds)	ball	FB	0.284	0.367 / 0.313 / 0.173
phase 1 (2 seeds)	ball	bounded, no reg.	**0.922**	0.940 / 0.903
phase 2 (2 seeds)	ball	visitation, ψ frozen	0.887	0.867 / 0.907

Mean over evaluations from 250k–500k steps, `cube-single-play-v0`. **The privileged goal region does not explain the gain — given to FB's own estimator it makes things worse**, 0.284 against the 0.420 anchor. The bounded residual is what makes a ball-shaped reward usable at all.

The mechanism is visible in the critic scale, and the arms are provably matched (shared seeds give identical initial values, 3.70 and 8.63):

arm	$\overline{\lvert M \rvert}$, start $\to$ 500k
control (FB loss, ball)	$3.70 \to 812$, $8.63 \to 1147$, $6.80 \to 1097$
phase 1 (bounded, ball)	$3.70 \to 0.187$, $8.63 \to 0.188$

FB's diagonal term is a *linear* — hence unbounded — attraction on the reward support. Widening that support from I (a 1/256 fraction of the batch matrix) to the ball (a measured 1.74%) puts roughly 4.5× more linear pull into the same loss, so it diverges *faster* than stock FB. This is the controlled version of the frozen-ψ experiment of Section 7.5: there we held the representation fixed and swapped the estimator; here we hold the estimator fixed and swap the reward support. Both point at the linear diagonal term, and together they narrow — though they still do not fully isolate, since the control varies support and not the linear term itself — the ablation the caveats section flags as missing.

7.7 Phase 2: the visitation measure

Protocol: initialize all networks from the matching phase-1 checkpoint, freeze ψ, drop the absorption mask from the backup (Algorithm 2), actor greedy term normalized by the centered magnitude. Both seeds complete 500k with no instability at any point: finals **0.952/0.928**, full curves in 0.89–0.99, and the critic scale plateaus at ≈ 24 against the Monte-Carlo visitation mass of 24.7 (Figure 5) — a $\sim$2% error in the measure's own units, obtained with no scale supervision. This is an *aggregate scale* check and nothing more: it compares one mean against one mean, and is by construction blind to how the mass is distributed over goals, and to its sign. Section 7.8 tests the same object pointwise and finds both of those to be wrong, so the 2% should not be read as evidence that the learned object is calibrated as a measure.

7.8 Is it a measure? A direct calibration test

Every headline number in this note scores an *actor*. That metric cannot detect whether $\tilde{M}$ is a measure, because the actor only ever consumes $\arg\max_a \langle F(s,z,a), z\rangle$: an ordering over actions at a single state. A critic can order actions perfectly while being an arbitrary function of g. Task success and measure validity are therefore logically independent, and the note has so far conflated "the curriculum trains stably to a good policy" with "the curriculum learns a successor measure." We now test the second claim on its own terms, on the phase-2 checkpoints of Section 7.6.

That the two come apart is already measurable on the JAX checkpoints. Over a grid of 24 states × 150 goals × 16 actions restricted to reachable (s, g) pairs, the reaching value and the visitation measure agree on the greedy action only

$$\Pr\left[\arg\max_a Q = \arg\max_a M\right] = 0.295 \ / \ 0.299 \quad \text{(2 seeds)},$$

with rank correlation between Q and M over actions of 0.40/0.38 (means). The actor is trained greedy on $\langle F, z\rangle$, i.e. on Q; on roughly seven of ten state–goal pairs it therefore does *not* choose the action its own occupancy measure prefers. A high success rate consequently certifies Q, not M. The same grid refutes the pointwise action-independence that Eq. 1 assumes of the recurrence factor — $\mathrm{ICC}(R \mid g) = 0.242/0.292$ against a pre-registered bar of >0.9, with within-goal coefficient of variation 1.35/1.75 — so the factorization is a useful decomposition of magnitudes, not an identity that survives conditioning on the action.

Protocol. For a start row $(s_0, \text{qpos}, \text{qvel})$ from the dataset, on-manifold $z = \psi(g)$ for a random dataset state, and $a_0 \sim \pi(s_0, z)$, compare

$$\underbrace{\hat{M}(X) = \tfrac{1}{n} \sum_{i:\, x_i \in X} \tilde{M}(c(x_i))/\hat{N}(c(x_i))}_{\text{model}} \quad \text{against} \quad \underbrace{\mathbb{E}\big[\textstyle\sum_k \gamma^k \mathbf{1}[\text{cube}_{1+k} \in X]\big]}_{\text{Monte Carlo, } \pi_z \text{ rolled out}}$$

over random boxes X at three half-widths, 32 start states × 3 rollouts, horizon 200 ($\gamma^H = 0.134$ of the mass is beyond the horizon and is reported, not hidden). The index convention matches the training residual: the sum starts at the state *after* a_0 with weight γ^0. We run the denominator two ways — exact KDTree kernel counts $\rho(B_\epsilon(g))$, which isolate whether $\tilde{M}$ itself is calibrated, and the trained $\hat{N}$ head, which scores the phase-2 object end to end.

A harness requirement, stated because we got it wrong. Teleport-style Monte Carlo of this kind must call `env.reset()` *before* every rollout, not merely overwrite `qpos`/`qvel`. The Gymnasium `TimeLimit` step counter otherwise carries across rollouts, so every rollout after the first truncates at step 1 and the estimated occupancy collapses to near zero, inflating every model/MC ratio by one to two orders of magnitude. An earlier version of this section reported ratios of 66–215× and concluded that the learned object was not a measure; that conclusion was an artifact of this bug and is withdrawn. The tell available at the time, and the check we now recommend, is a sanity bound on the denominator itself: the implied total Monte Carlo mass was ≈1–3 against $\sum_k \gamma^k \approx 87$, which is impossible, since the cube is always somewhere. All numbers below come from the corrected harness.

		model / MC		
half-width	MC	exact ρ	trained $\hat{N}$	Spearman
0.4	0.56/0.86	3.22/2.03	1.94/1.22	+0.31/ + 0.33
0.8	5.98/5.82	1.92/2.08	**1.10/1.16**	+0.68/ + 0.72
1.6	27.4/28.1	2.33/2.18	**1.23/1.14**	+0.85/ + 0.82

At the two larger scales the end-to-end object is calibrated to within 10–23%, and multiplying by the truncation factor $1 - \gamma^H = 0.87$ — the model estimates an infinite-horizon mass while the Monte Carlo sum stops at H=200 — brings the ratio to ≈1.0. The pre-registered gate for licensing $\int r\, dM$ was a relative error of order 15%, and at half-widths 0.8 and 1.6 that gate is met. Rank correlation is +0.68 to +0.85 there. *At set level, on this environment, the phase-2 object behaves like a measure.*

Three qualifications keep this from being stronger than it is. First, the calibrated column is the one that divides by the *trained* $\hat{N}$; with exact kernel counts the same object is 1.9–3.2× off. The head over-predicts density by +28% (below), which deflates the ratio, so part of the agreement is two biases cancelling: read the $\hat{N}$ column as what the method delivers end to end, and the exact column as how wrong $\tilde{M}$ itself is. Second, calibration degrades at the smallest scale — 1.2–1.9× and Spearman ≈0.32 at half-width 0.4, which is the radius of the training kernel itself, so the object is worst at exactly the resolution it was fit at. Third, all of this is one environment.

The object takes negative values. Over 8 draws of (s_0, a_0, z) per seed and 100k goal states each,

$$\Pr[\tilde{M} < 0] = 45.0\% \text{ (seed 42)}, \qquad 45.9\% \text{ (seed 43)},$$

per-draw range 7.1%–55.2%; over 200k goals $\tilde{M}$ has mean 1.92, standard deviation 7.57, and range [−12.17, +47.36]. A successor measure is non-negative by definition, so pointwise the learned object is not one, even where its set-level mass is right: the negative entries cancel against positive ones inside a box, which is why aggregate calibration can look good while the allocation underneath is wrong. Nothing in the training constrains the sign — the bounded residual bounds the *fixed point*, while the bilinear form $F^\top \psi$ remains free to go negative wherever the regression is not pinned by data.

The obvious repair trades sign against scale. Adding a one-sided squared hinge $\lambda\, \mathbb{E}[\text{relu}(-\tilde{M})^2]$ — a penalty rather than a softplus or squared parameterization, so that $\tilde{M}$ stays linear in ψ and the $\langle F, z_r \rangle$ readout survives — does remove the negative mass: 45% → 3.1% at λ=1 and → 0.7% at λ=10, at

no cost in task success. But it inflates set mass monotonically in λ, to 2.5–3.6× at half-width 0.8 against the unpenalized 1.10/1.16. The hinge pushes negative entries up with nothing conserving total mass. We therefore do not adopt it: a mass-conserving non-negativity constraint remains open, and is the clearest single gap in the object.

The denominator half-worked. The phase-2b head $\hat{N}(g)$, scored against exact kernel counts on 4k goals, attains rank correlation 0.994 but median relative error 23.7% and a +29% bias (0.0171 vs. 0.0133). Since $\hat{N}$ is the *easy* half — pure counting, no bootstrapping — its residual error bounds how good the ratio can be even if $\tilde{M}$ were fixed.

This sharpens, and makes quantitative, the qualitative statement in Section 11 that the diagonal critic "collapses to ≈ 0 or below" away from its hindsight support: the collapse is not a far-field curiosity but covers close to half of the natural (s, g) product, and it is signed. The net reading is a split verdict — set-level mass and ordering at the coarser scales are licensed, pointwise values are not — and the practical question is which of the two the downstream readout actually needs. Section 7.9 answers that empirically by running novel rewards through it. The old blanket statement that set-mass readouts are unlicensed does not survive the corrected harness; what does survive is that a mass-conserving route to non-negativity remains a genuine prerequisite for trusting $\tilde{M}$ pointwise, rather than a refinement of it.

7.9 Novel-reward transfer: construction, baselines, and a bandwidth limit

Section 7.8 leaves a split verdict: set-level mass and coarse ordering are calibrated, pointwise values are not. Which of the two a user needs depends on the readout, so we test the readout directly, on rewards the model has never seen. This is the capability the whole programme is for — if a successor measure is worth having, an unseen reward should be servable without training.

Reward construction. All rewards are functions of the cube position $c(x) \in \mathbb{R}^3$ (observation slice `[19:22]`, in observation units where $1 = 10\,\text{cm}$, so the trained kernel radius $\epsilon = 0.4$). Centers are drawn uniformly from dataset cube positions, so every reward is supported on visited space. Six families × 8 instances = 48 rewards, none seen in training:

Family	Definition	Relation to the trained kernel
BALL 0.4	$\mathbf{1}[\|c(x) - c_0\| \le 0.4]$	radius *equals* ϵ; easiest case
BALL 0.8	$\mathbf{1}[\|c(x) - c_0\| \le 0.8]$	2× the kernel
BALL 1.6	$\mathbf{1}[\|c(x) - c_0\| \le 1.6]$	4×; a 16 cm ball
SMOOTH 0.4	$\exp(-\|c(x) - c_0\|^2/2\sigma^2)$, σ=0.4	no threshold, unbounded support
SMOOTH 0.8	as above, σ=0.8	widest support tested
TWOZONE	$\max_k \mathbf{1}[\|c(x) - c_k\| \le 0.4]$, $\|c_1 - c_2\| > 1.6$	tests *composition*, not reach

The family axis is deliberately a *bandwidth* axis: it sweeps the reward's spatial support from the kernel's own radius outward, which is the axis on which a measure built from ϵ-balls might be expected to fail.

Inference and the two stages. A task vector is inferred by the standard moment readout on m=10,000 relabeled dataset rows, $z_r = \Pi\big(\sum_i r(x_i)\,\psi(x_i)\big)$, with no ridge, no $\hat{N}$ correction and no GPI — the minimal pipeline, so that what is tested is the representation rather than the decoder. Stage A scores *prediction*: from 12 teleported starts × 2 rollouts we compare $\langle F(s_0, z_r, a_0), z_r\rangle$ against the Monte Carlo discounted return of $\pi(\cdot, z_r)$ under the new reward. Stage B scores *control*: 10 episodes per drive at horizon 200, $\gamma = 0.99$.

Baselines. Stage B reports a normalized gap $\text{gap} = (G_{z_r} - G_{\text{floor}})/(G_{\text{ceil}} - G_{\text{floor}})$ against three references, which we state explicitly because the choice of floor changes the reading: *ceiling* z_{goal}, the projected ψ of the dataset state nearest the reward's center (for TWOZONE, the better of the two centers) — an oracle handed the answer, and a *proxy* for the true optimum, since "go to the center" is not optimal for a broad Gaussian; *floor* z_{rand}, a random task vector driving the same actor; and the *behavior* baseline, the discounted return the offline data itself earns on each reward (1000 episodes, same horizon and discount). The last is the one a reader should anchor on, and it was added late: on the concentrated families it agrees closely with z_{rand} (1.21 vs. 0.91 on BALL 0.4; 2.04 vs. 2.09 on TWOZONE), but on the

broad families it does not — on BALL 1.6 the offline behavior earns 25.1 while a *random* task vector earns 35.6, because a 16 cm ball rewards an actor that merely moves the cube around. Normalizing against z_{rand} therefore understates broad-reward performance, and we report both.

	Stage A: prediction		**Stage B: control**		
Family	Spearman	scale	gap vs. z_{rand}	gap vs. behavior	G_{z_r} / ceil
BALL 0.4	0.85/0.90	0.014	0.87/0.74	0.80	15.6 / 19.1
BALL 0.8	0.80/0.81	0.023	0.70/0.61	0.65	21.8 / 30.1
TWOZONE	0.73/0.69	0.017	**0.83/0.89**	**0.86**	15.8 / 18.1
SMOOTH 0.4	0.75/0.86	0.019	0.54/0.46	0.52	11.4 / 19.1
SMOOTH 0.8	0.62/0.75	0.018	**0.23/0.08**	0.32	19.4 / 34.2
BALL 1.6	0.72/0.81	0.018	**−0.10/0.24**	0.44	37.5 / 53.1

Table 3: Zero-shot transfer to 48 never-trained rewards on the λ=0 phase-2 checkpoints, eval-only. Stage-A Spearman per seed; scale is $\mathbb{E}[\text{pred}]/\mathbb{E}[\text{MC}]$, target 1. Stage-B gaps per seed against z_{rand}, and 2-seed mean against the behavior baseline.

Three readings. (i) *Transfer works, and composes*: concentrated rewards reach 0.66–0.86 of the oracle, and TWOZONE — two disjoint targets, which no single goal can express — is the *strongest* family. (ii) *Ordering is uniformly good, scale uniformly wrong*: Spearman stays 0.62–0.90 across every family, including ones where control collapses, while the predicted magnitude is 0.014–0.030 of the true return in every family. That band is not free: the mean kernel mass measured independently in §7.8 is $\bar{N} = 0.0137$, which is the $1/\hat{N}$ density division the naive moment readout omits — exact on BALL 0.4 (0.0136) and within a factor of two on the rest. Re-running inference with weights $1/\hat{N}(c(x_i))$ moves the scale to 0.79–2.2, confirming the diagnosis quantitatively. (iii) *But the density correction does not transfer to behavior*: the same weighting leaves control unimproved everywhere and *regresses* it where it worked (TWOZONE 0.86 → 0.46). Prediction and control therefore want different readouts — naive z_r for acting, $1/\hat{N}$-weighted for integrating — which is a symptom rather than a design, and localizes the residual defect away from the inference rule and onto the actor: a broad reward's z_r averages many ψ's and lands off the manifold the actor was trained on, which is 50% random directions and 50% ψ of a single goal.

7.10 Populating the task-vector manifold

That diagnosis is testable by changing the training z distribution and nothing else. We add a third component to the z law, drawn exactly as the zero-shot readout draws it — $z = \Pi(\sum_i r_i \psi(x_i))$ for a random synthetic reward r over the batch's next states — at mixing fraction η, keeping phase 1, the objective, and every other hyperparameter fixed, and warm-starting from the same phase-1 checkpoints as the control. Two reward generators, deliberately distinguished: KERNEL draws balls, Gaussians and two-zone rewards with random centers from the batch and log-uniform radii in $[0.2, 2.0]$ — the eval families' *functional form*, so it upper-bounds what the actor can learn; RFF draws $r = \text{relu}(\cos(w^\top c + \varphi))$ with $w \sim \mathcal{N}(0, I)/b$ — non-negative random fields of controlled length scale, a *different* functional form, so the evaluation families remain held out and a win cannot be an artifact of training on the test distribution.

The pre-registered primary read was that the broad families rise above 0.4 while the concentrated ones hold above 0.7. It is half met, in an informative way. BALL 1.6 clears decisively — $-0.10/0.24 \rightarrow 0.47$–$0.55$ on every arm and both seeds — while SMOOTH 0.8 does not, topping out at 0.33. Broad *indicator* rewards are fixed; broad *Gaussian* rewards are not, and the difference between them is that the Gaussian has no support boundary at all, so its z_r is an average over essentially the whole dataset. Both guards pass: task success does not degrade but improves ($0.887 \rightarrow 0.93$–0.95, ≈2 seed-sd, 2 seeds so suggestive only), and set-mass calibration is unchanged.

Two structural points. First, the mechanism is confirmed: because the *matched-form* arm works, the actor's z-manifold really was the binding constraint, as §7.9 inferred rather than showed. Second, the result is not circular: RFF, whose training rewards share no functional form with any evaluation family, matches or beats the matched arms on BALL 1.6. What the z law supplies is coverage of the broad-z

Family	control	KERNEL η=0.25	KERNEL η=0.5	RFF η=0.25
BALL 0.4	0.80	0.87	0.87	0.79
BALL 0.8	0.66	0.71	**0.95**	0.73
TWOZONE	0.86	0.92	**1.35**	1.11
SMOOTH 0.4	0.50	0.49	**0.78**	0.55
SMOOTH 0.8	0.15	0.27	0.33	0.12
BALL 1.6	**0.07**	**0.53**	**0.49**	**0.55**
mean (all 6)	0.507	0.630	**0.795**	0.640
task success	0.887	0.948	0.950	0.930
$\hat{N}$ ratio @ hw 0.8	1.10/1.16	1.10/1.19	1.23/1.31	1.26/1.12

Table 4: Stage-B gap vs. z_{rand}, 2-seed means, with the two pre-registered guards. Thresholds were fixed before the runs.

region, not familiarity with the test distribution. Finally, TWOZONE at 1.35 exceeds the single-goal oracle ceiling by 35% — as it should, since that ceiling is the better of the two zones and a genuine measure can serve both.

7.11 Scope: the manipulation environments with a published FB number

Sections 7.4–7.7 are all `cube-single`. Table 7.11 extends both phases to the four OGBench state-based manipulation environments for which a tuned vanilla-FB number exists in the literature (Bagatella et al., 2025) — exactly the set that reference sweeps (`cube-single`, `cube-double`, `scene`, `puzzle-3x3`). The recipe is held fixed at the §7.4 winner; the *only* per-environment quantity is the goal region, which in each case mirrors that environment's own success criterion (a conjunction over objects: both cubes; cube ∧ two buttons ∧ drawer ∧ window; all nine button states).

	this note (500k)		**anchor (500k)**	**OGBench reference (1M)**	
Environment	phase 1	phase 2	GC-IQL bilinear d=50	GC-IQL	HIQL
`cube-single`	**0.967**	0.955	0.928	0.68 ± 0.06	0.15 ± 0.03
`cube-double`	**0.549**	0.358[b]	0.265	0.40 ± 0.05	0.06 ± 0.02
`scene`	**0.484**	0.342[b]	0.281	0.51 ± 0.04	0.38 ± 0.03
`puzzle-3x3`	**0.200**[e]	0.198[e]	0.117[d,e]	0.95 ± 0.01[d]	0.12 ± 0.02

Overall success (mean over the five native goal-conditioned tasks), $[0, 1]$. **The two halves of this table are not the same experiment** and should not be read as a like-for-like ranking. Ours: 500k gradient steps, 2–3 seeds, the flow-BC+distillation actor throughout. OGBench (Park et al., 2024): 1M steps, 8 seeds, the stock reference implementations (reported there as percentages; divided by 100 here). The *anchor* column is our own GC-IQL run — bilinear d=50 critic and the same flow actor as the measure cells — and is the only column matched to ours in budget, actor class, and critic rank; it is what the phase columns should be read against. All cells are now complete at 500k: 3 seeds for phase 1 and the anchor, 2 seeds for phase 2 (3 on `cube-single`); per-seed values are given in the text where they matter. [b] phase 2 *loses* ground here — see below. [e] **degenerate cell.** On `puzzle-3x3` *every* run of *every* method scores exactly 0.0 on tasks 2–5; the entire column is task 1. All three phase-1 seeds score exactly 1.0 there (which is why they read 0.200 identically), phase 2 scores 1.0/0.98, and the anchor 0.28/0.74/0.74. The ∼2× ratio is therefore one task's margin, not a spread over five, and no method here solves any of the other four. The row should be read as "all methods fail" with a tie-break, not as a phase-1 win. [d] **unresolved.** Stock GC-IQL is reported at 0.95 on `puzzle-3x3` while our bilinear-d50 anchor reaches 0.117 — an eightfold gap that no other row shows, and the one number in this table we would not yet cite. The likeliest cause is per-environment tuning: our anchor carries `bc_coeff`=3.0, selected on `cube-single`, to every environment unchanged. Until that is separated from an architectural effect, the `puzzle-3x3` anchor should be treated as un-tuned rather than as a measurement of what GC-IQL can do there.

Three readings, and one thing the table does not say.

Phase 1 exceeds its matched anchor on all four environments, by 0.04 on `cube-single` (where the anchor is already near ceiling), 2.1× on `cube-double`, 1.7× on `scene`, and nominally 1.7× on `puzzle-3x3` — though that last is the degenerate cell of footnote e and should not be counted. Since the

anchor shares the data, the actor class, the critic rank and the training budget, the reaching objective — not the representation or the extraction — is what differs. The lead caveat of this note therefore weakens from "one environment" to "four environments, one benchmark family, state observations"; it does not vanish.

Phase 2 does not hold the phase-1 level once off `cube-single`. The visitation phase costs 0.549 → 0.358 on `cube-double` (−0.19) and 0.484 → 0.342 on `scene` (−0.14), while `cube-single` is flat (0.967 → 0.955) and `puzzle-3x3` is flat because it is degenerate. Phase 2 still clears the anchor everywhere, so the curriculum is not broken — but the earlier reading that the visitation phase is free, drawn when `cube-single` was the only completed column, does not survive the other three. Together with Section 7.8, where the phase-2 object fails its calibration gate outright, the honest summary is that **phase 2 currently costs success and does not buy a usable measure**; its justification rests on the semantics it is supposed to provide, which are not yet demonstrated.

The absolute numbers fall away from `cube-single` regardless of method. Phase 1 drops 0.97 → 0.55 → 0.48 → 0.20 as the goal region becomes a conjunction over more objects, and the anchor drops with it. Whether that is the conjunction itself, the resulting sparsity of the hindsight matrix (measured off-diagonal positives per row at batch 1024: 13.9, 0.46, 2.85, 2.04 respectively), or simply that these environments are harder, *this table* cannot separate — the candidate explanations are confounded across the four environments.

The sparsity limb, however, is separable within a single environment, and each cell ships such a control: hold the environment and every hyperparameter fixed and drop objects from the goal-region conjunction, which *widens* the region and therefore *densifies* the hindsight matrix.

environment	conjunctive region	single-object control	ratio
`cube-double` (both cubes vs. one)	**0.549**	0.150	3.7×
`scene` (full conjunction vs. cube only)	**0.484**	0.228 (0.224/0.232)	2.1×

The denser cell loses in both environments, and on `cube-double` the region that is ∼30× sparser wins by 3.7×. So the pre-registered sparsity reading — more hindsight positives per row should train better — is *refuted*: whatever phase 1 exploits, it is not positive density in the training matrix, and the fall-away across Table 7.11 is not explained by it. What remains confounded is the conjunction itself against raw environment difficulty, which these controls do not address.

What the table omits is the vanilla-FB comparison that motivated this environment set. FB is reported at 0.496/0.026/0.128/0.048 on these four (Bagatella et al., 2025), far below every column above — but under a *different evaluation protocol*: a task is specified by reward labels and z is inferred from 10^4 relabelled samples, with 10 episodes per task, whereas every number in Table 7.11 is native goal-conditioned evaluation with the goal handed to the policy. The latter is the easier setting. Our protocol-matched numbers are the zero-shot rows of Table 7.12, which exist only for `cube-single`.

Extending them does not require porting the JAX reward-family harness off `cube-single` after all: the torch re-implementation of Section 7.6 already carries FB's own zero-shot harness, so running the curriculum there supplies protocol-matched numbers directly. Both phases have now been trained in that codebase on `scene` and `cube-double` (500k steps, phase 2 warm-started from the matching phase-1 checkpoint with B frozen and the $\hat{N}$ head on; *one seed*, and checkpoints only — **no evaluations have been run on them**). Until those are scored and seeded we make no head-to-head FB claim on `cube-double`, `scene`, or `puzzle-3x3`.

7.12 Zero-shot extraction: every family, every route

Table 7.12 reports the complete zero-shot grid: four inference routes × six reward families, per seed. Routes: *naive* (moment z_r + goal-proxy decode, on phase-1 checkpoints — run before we understood that inference must use the critic's embedding space, and kept as the historical control); *retrieval-1* on phase-1 and on phase-2 checkpoints (top-1 reward-weighted state as goal, GPI over candidates); and *direct-z* on the separated-form checkpoints (§7.13; z_r consumed by F and π with no decoding).

Family	naive (P1)	retrieval-1 (P1)	retrieval-1 (P2)	direct-z (sep.)
DIRAC	0.42/0.41/0.57	0.89/0.96/0.97	0.94/0.76	0.90/0.92
MIXTURE (10 pairs)	0.83/0.68/0.64	0.86/0.74	0.77/0.80	0.87/0.89
REGION	0.57/0.65/0.29	0.80/0.74	0.73/0.73	0.61/0.64
DENSE	0.34/0.46/0.20	0.99/0.98	0.93/0.93	0.90/**0.13**
REACH-AVOID (succ. at A)	0.00×3	0.96/0.91	0.66/0.90	0.33/0.00
LINEAR (z_r vs. control)	mixed[a]	11/−10; 19/−67	mixed	≈ no edge

Per-seed values separated by "/"; retrieval-1 (P1) has 3 seeds for DIRAC and 2 for the remaining families (seed 0 ran DIRAC only). Fraction of time in the avoid region B is ≤ 0.003 in every cell, including ones that never reach A, so the avoid side of REACH-AVOID is not yet discriminative. [a] achieved discounted reward, z_r vs. a random-direction control: 73 vs. 85, 5 vs. −9, 49 vs. −68 across phase-1 seeds — 2/3 beat the control by large margins with large variance; we treat LINEAR as unresolved rather than null.

Secondary sweeps. *Retrieval size*: k=1 dominates (0.89/0.96/0.97 on DIRAC); k=4: 0.70 (seed 0); k=16: 0.90/0.54/0.78 — averaging embeddings over a larger retrieved set moves z off the trained manifold and is strictly less stable. *Recurrence-weighted inference*: weighting reward samples by the fitted $R(g)$ (the occupancy-correct integrand) changes DIRAC by < 0.01 — structurally validated, behaviorally inert on these goal-reducible families, as expected since they never query revisit mass.

Reading the grid: every family whose optimum reduces to one dataset state (DIRAC, single-peak DENSE, and to a lesser degree REGION and REACH-AVOID's reach side) is solved by retrieval at 0.73–0.99 — a property of the decode. The routes that consume z genuinely (naive, direct-z) are the ones that test the representation, and there the separated form closes most of the gap the naive route exposed (0.42 → 0.90 on DIRAC) while leaving two honest open items: the DENSE seed split and LINEAR.

7.13 The separated form: training and direct consumption

Embedding-conditioned actor control. The minimal change from the phase-1 winner: condition the actor on $z = \text{sg}[\psi(g)]$ (the critic's embedding) instead of the raw goal, nothing else. Result: 0.960/0.996 at 500k — at/above the goal-conditioned route. This isolates the earlier embedding-actor failures (0.395; collapse) to the task-sampling priors they were trained under, not to conditioning.

Separated training. The full form of §10: rows carry independent policy indices z_i (goal embeddings mixed 1:1 with random unit directions scaled to the data norm; bootstrap actions from $\pi(\cdot|s', z_i)$), columns carry the query; cold start. Both seeds train stably to **0.904/0.960** at 500k with critic scale bounded near 1 throughout; the mid-training collapse mode of the scalar-grounded variant (Table 7.3) does not appear. Grounding the task family through the matrix residual, rather than through an added scalar TD, is evidently the stable form.

Direct-z extraction is column 4 of Table 7.12: single goals 0.90/0.92 with $\cos(z_r, \psi(g^*)) \approx 0.96$, mixtures 0.87/0.89 — and DENSE at 0.90 vs. 0.13, a hard seed split. On the failing seed the same checkpoint scores 0.92 on DIRAC, so the actor and critic are intact; the dense-reward *inference* misaims. The suspected cause was covariance anisotropy (the moment readout equals the least-squares solution only under near-isotropic features; a dense reward's z_r averages hundreds of embedding directions and is far more sensitive to the local geometry of ψ than a point-goal's).

That diagnosis has since been tested and refuted. Replacing the moment readout with its ridge-regularized least-squares variant, at $\lambda \in \{10^{-3}, 10^{-2}, 10^{-1}\}$, gives

seed	moment	λ=10^{-3}	λ=10^{-2}	λ=10^{-1}
s0 (good)	**0.896**	0.472	0.512	0.648
s1 (bad)	0.128	0.432	**0.464**	0.432

Ridge rescues the failing seed only partially (0.128 → 0.464) and *destroys* the working one (0.896 → 0.472), at every λ tried. Anisotropy is therefore at most half the story, and whatever separates the two seeds is not a conditioning problem that a better linear solve fixes. The dense split stays open, with its leading explanation now eliminated rather than pending.

7.14 Negative results kept for the record

For completeness, cells that ran and died, with their diagnosis where we have one: the actor-side scale-centering fix on the "identity corner" cell (the configuration that is provably an affine reparameterization of GC-IQL) remains at 0.000 through 350k — the anomaly that the theorem-equivalent cell does not train is real and unexplained; task-sampling priors over the actor's z (Dirac/mixture/random arms) underperform from the start (0.395); and adding a scalar task-conditioned TD grounding to the critic produces the tracked-then-collapsed pattern on all 7 seeds.

Three readings we currently trust: (i) the reaching-value objective is learnable and controllable on this domain at the level of the goal-conditioned TD baseline; (ii) the orthonormality regularizer and the density-style linear TD term are each independently harmful here — the divergence occurs even with a perfect frozen goal embedding, so it is a property of the estimator, not of representation quality; (iii) replacing the density estimator by the bounded visitation regression removes the divergence while keeping genuine occupancy semantics (revisits counted) — both seeds completed 500k steps without instability, at the correct occupancy scale.

8 Policies extractable from the learned measure

At test time a task is specified by reward-labeled samples $\{(s_j, r_j)\}$ drawn from the dataset (relabeled offline, no interaction). The measure supports the following extraction routes today; success rates are on the frozen phase-1 checkpoints unless noted. The completed phase-2 checkpoints evaluate comparably through the same routes (single goals 0.94/0.76, mixtures 0.77/0.80, dense 0.93/0.93 across the two seeds) — capability is preserved, but no advantage over phase 1 appears on these goal-reducible families, which is expected: the families below never query the revisit mass that phase 2 adds.

1. **Goal-conditioned (trained route).** The actor conditioned on a goal state, greedy on the measure readout: 0.95–0.98.

2. **Single goal, zero-shot from reward samples.** Embedding-space inference ($z_r \propto \sum_j r_j \psi(s_j)$, actor conditioned on a decoded goal) is fragile when decoded naively: 0.41–0.57, bimodal per task. Decoding by *retrieval* — take the top-1 reward-weighted dataset state as the goal — recovers 0.888/0.960/0.968 (3 seeds). Larger retrieval sets are less stable (k=16: 0.54–0.90).

3. **Goal mixtures ("reach either").** Generalized policy improvement (Barreto et al., 2017) over the candidate goals — act with the candidate whose readout is highest in the current state. Over all 10 task pairs the family means are 0.64–0.89 per seed/route (Table 7.12); the easiest pairs reach 0.94–1.00, the hardest (involving the stacking tasks) drag the mean.

4. **Dense rewards with a single optimum also work — because they reduce to a goal.** A shaped reward $r(g) = \exp(-\|\text{cube}(g) - c^*\|/\sigma)$ evaluates at 0.93–0.99 under the same retrieval decode (phase-1 checkpoints 0.976/0.992; phase-2 checkpoints 0.93/0.93). This is a property of the *decode*, not of the measure: the reward's argmax is a single dataset state, so the task is goal-reducible. Under the naive embedding decode the same family sat near chance — the earlier failure was route-specific.

5. **What does not work yet.** Rewards that do *not* reduce to a small candidate-goal set. Concretely (Table 7.12): REGION tops out at 0.73–0.80 even with retrieval (below every goal-reducible family); LINEAR rewards are unresolved — z_r beats a random-direction control on 2/3 phase-1 seeds but with enormous variance and no edge on the separated form; and multi-region compromises/penalty structures have no working route (the avoid side of REACH-AVOID is not yet discriminative — time-in-B is ≈ 0 for every policy). Section 9 explains why this boundary is structural rather than an actor-decoding inconvenience. One earlier obstacle has since been removed: conditioning the actor on the task embedding $z = \psi(g)$ *instead of* the raw goal — the form a reward-embedding-consuming policy needs — trains cleanly on the phase-1 objective (0.960/0.996 at 500k, 2 seeds), matching the

goal-conditioned route. Earlier failures of embedding-conditioned actors are now attributed to the task-sampling priors they were trained under, not to embedding-conditioning itself.

Task inference: how z is extracted, intuitively and formally. A task arrives as reward-labeled dataset samples $\{(x_j, r_j)\}_{j=1}^m$. The goal embedding ψ doubles as a feature map for rewards: if a reward is (close to) linear in the features, $r(x) \approx \psi(x)^\top z_r/\sqrt{d}$, then the critic slice at z_r *is* that reward's action-value, so inference is a linear regression in embedding space:

$$z_r \;=\; \arg\min_z \sum_j \big(r_j - \psi(x_j)^\top z/\sqrt{d}\big)^2 \;\;\propto\;\; \Big(\tfrac{1}{m}\sum_j \psi(x_j)\psi(x_j)^\top\Big)^{-1} \tfrac{1}{m}\sum_j r_j\,\psi(x_j),$$

which reduces to the FB moment readout $z_r \propto \mathbb{E}[r\,\psi]$ when the feature covariance is near-isotropic (Touati and Ollivier, 2021). We rescale z_r to the typical embedding norm of the data (the covariance of a regularizer-free ψ is anisotropic, so the raw magnitude is uninformative), and the inferred vector is consumed by *both* heads of the separated form: the critic $F(s, a, z_r)$ and the policy $\pi(a \mid s, z_r)$ — no goal decoding. Intuition for why this works after phase 1: the reaching objective trains ψ so that a goal's indicator reward is (approximately) a one-dimensional direction in embedding space; a single-goal reward therefore infers to $z_r \approx \psi(g^*)$ — measured cosine ≈ 0.96 on the separated-form checkpoints — and a mixture infers to a weighted combination of its goals' directions. One practical caveat discovered the hard way: the inference must run in the embedding space the actor was trained against (here the *critic's* ψ; measure-only training leaves any auxiliary value network untrained, and using its random ψ silently degrades every embedding-space number).

Direct-z extraction results (separated-form checkpoints, 2 seeds). Consuming z_r directly, with GPI over $\{z_r\} \cup \{\psi(g_k)\}$ candidates: single goals 0.90/0.92; two-goal mixtures 0.87/0.89; dense single-optimum rewards 0.90/0.13. The dense split (one seed near anchor, one near floor) means dense-reward inference is not yet robust — under investigation — while single-goal and mixture extraction look stable. For calibration, the naive embedding route on the goal-indexed checkpoints sat at 0.41–0.57; the gap closed by (i) the separated training form and (ii) inferring in the correct embedding space.

9 What object is actually learned, and what it licenses

The critic of both phases has signature $(s, a, g) \mapsto \phi(s, a)^\top \psi(g)$, and the bootstrap actions in the TD target are drawn from $\pi(\cdot \mid s', g)$ — the policy *of the same* g. So what phase 2 learns is

$$g \;\longmapsto\; M^{\pi_g}\big(s, a, B_\epsilon(g)\big),$$

the visitation value of each goal region *under that goal's own continuation policy*: the diagonal slice of the two-argument object $(z, x) \mapsto M^{\pi_z}(s, a, B_\epsilon(x))$ that successor-measure zero-shot RL requires.

This distinction — goal-dependent values versus a fixed-policy measure over all futures — is classical (Blier et al., 2021), and it predicts our empirical pattern exactly. Summing the slice over two rewarded goals combines $M^{\pi_{g_1}}(\cdot, g_1)$ and $M^{\pi_{g_2}}(\cdot, g_2)$ — quantities computed under *different* continuation policies — so the sum is not the value of any single policy; it can rank a small candidate set (that is GPI, and it works) but cannot value a compromise between rewarded regions (dense rewards, and they fail). A second, independent gap: even under one policy, the ball value $M(s, a, B_\epsilon(g))$ is a *kernel-smoothed mass*, not a density ratio with respect to the data distribution. Integrating it over goal centers double-counts overlapping balls and inherits the dataset density over g; writing $Q_r = \int r\, dM$ for the learned object is therefore not yet justified without kernel normalization or an importance correction. Until both gaps are closed, the honest description of Sections 4–5 is: *stable goal-conditioned visitation learning with strong goal-policy composition*, not arbitrary-reward zero-shot RL.

10 Toward dense rewards: separating the policy index from the query

The repair is to give the critic and the policy the two arguments the full object has:

$$M(s, a, z, x) = F(s, a, z)^\top B(x), \qquad a' \sim \pi(\cdot \mid s', z),$$

with TD target

$$y_{ijk} \;=\; h(s'_i, x_k) \;+\; \gamma\, F(s'_i, a'_i, z_j)^\top B(x_k), \qquad a'_i \sim \pi(\cdot \mid s'_i, z_j)\,:$$

rows range over transitions, z_j over sampled policy indices (goal embeddings mixed with random directions, so the family is trained where inferred task vectors live), x_k over query regions — and crucially z and x vary independently. At test time $z_r = \mathbb{E}_{x\sim\rho}[r(x)B(x)]$ is consumed *directly* by both F and π, with no goal decoding.

The composition's first stage now has evidence: training the goal-*reaching* residual in exactly this separated form — rows carrying independent policy indices z_i (goal embeddings mixed 1:1 with random directions, bootstrap actions from $\pi(\cdot|s', z_i)$), columns carrying the query, and a z-conditioned actor — is stable from a cold start and reaches the anchor: 0.904/0.960 at 500k (2 seeds), with the critic scale bounded throughout. Still open: the non-absorbing (visitation) phase on top of this form, and the test that actually matters — extraction quality when z_r is inferred from non-goal-reducible rewards and consumed directly, with no goal decoding. Until that test passes, the dense-reward column of the zero-shot table is claimed only for goal-reducible rewards, and this note claims nothing beyond it.

11 A benchmark, and a first full measure

The capability boundary of Sections 9–10 was, until recently, argued from theory and scattered reward families. It is now *measured*, by a frozen evaluation suite (SMB; spec 11) with two wings. **Wing A** extends the zero-shot control protocol with the tiers that discriminate objects rather than decodes: an *occupancy-weighted choice* between a stable (on-table) and a transient (elevated) rewarded region whose weights are placed between the reaching ratio and the occupancy ratio — so any method whose object is $\mathbb{E}[\gamma^T]$ selects the wrong region by construction (dataset recurrence gap 8.8–9.8×, all pairs validated before any checkpoint was scored); a corridor reach-avoid whose penalty region provably binds; unions; and the feature-linear family. **Wing B** evaluates the measure *as a measure*: predicted discounted occupancies against the empirical visitation of the checkpoint's own rollouts (episodes not terminated at goal), over a frozen battery of balls, boxes, and unions — including an additivity check — plus the calibration of $\int r\, dM$ against realized returns. A survey of the FB/successor-measure literature (12 papers) found no prior quantitative instance of any Wing-B metric; published evaluations route exclusively through rollout reward.

Three measured facts frame what follows. The GC baseline fails exactly where predicted: at chance on the occupancy choice, and structurally miscalibrated in Wing B (its readout is an affine reaching value; predictions of -40 against positive occupancies). The *diagonal* phase-2 critic is occupancy-calibrated only near its hindsight support: at dataset states its own-successor readout sits at the dwell scale (median ≈ 96), while its far field — including episode-start states — collapses to ≈ 0 or below; it is a proximity-calibrated object, not a measure over the (s, g) product. The separated (z,x) phase-2 is the best-calibrated row throughout, consistent with random-z training spreading the support.

The eval-layer full measure. The separated checkpoint's readout at a dataset state x *is* the kernel-smoothed mass $\tilde{M}(s, a, z, B_\epsilon(x))$, and the dataset supplies the same kernel's data mass $\hat{N}(x) = \hat{\rho}(B_\epsilon(x))$ as a pure count. The ratio and its integrals promote the slice to a measure with no retraining:

$$\hat{m}(s, a, z, x) = \frac{\tilde{M}(s, a, z, B_\epsilon(x))}{\hat{N}(x)}, \qquad \hat{M}(s, a, z, X) = \frac{1}{n} \sum_{i:\, x_i \in X} \hat{m}(s, a, z, x_i), \qquad \hat{Q}_r(s, a, z) = \frac{1}{m} \sum_j r_j\, \hat{m}(s, a, z, g_j),$$

with set masses finitely additive *by construction* and the same-kernel smoothing (including clipped-ball boundary effects at the table) cancelling in the ratio. Inference follows: the moment readout acquires the weight $1/\hat{N}(g_j)$, and the task vector is *selected* by ranking candidates with $\hat{Q}_r$ — the measure evaluating its own policies — rather than trusted open-loop. One numerical guard is required: at thin-support regions the $1/\hat{N}$ weight amplifies critic over-smoothing (a classical importance-weight tail), so cross-region comparisons winsorize the density's upper tail; capping at the 80th percentile saturates (the 60th changes nothing), which locates the *remaining* choice error in the critic's recurrence resolution, not the inference layer.

Gate results (two seeds; ablations isolate each component; the phase-1 control confirms the density layer cannot rescue reaching units, so phase-2 visitation training is what carries the occupancy content): the feature-linear family is *resolved* — the full route beats both the random-z and best-candidate controls on both seeds — the first family beyond goal-reduction to clear its controls; the union additivity defect drops $0.96 \to 0.19$; $\int r\, dM$ calibration improves from a $25\times$ scale error to slope 0.35–0.41 ($\rho \approx 0.66$); own-ball prediction error halves; and measure-ranked selection eliminates the signed-reward failure mode in which inferred z corrupted by negative weights destroyed the task (success under penalties retained at 0.77–0.83 against a collapse to 0.02–0.24 without it), though genuine avoidance (detouring) has not yet appeared. On the occupancy choice the winsorized route reaches 0.74/0.93 of the oracle return, above the un-corrected baseline on both seeds, with the choice rate at 0.58/0.65. Still open, with located causes: box queries (the kernel-average surrogate is the bottleneck; trained region columns or an additive region embedding are the designed fix), the residual choice error (critic recurrence resolution at thin support), and avoidance (requires off-manifold actor competence, never trained).

12 Caveats and relation to prior work

Caveats: four environments so far, all OGBench manipulation with state observations (§7.11); phase 2 is complete only on `cube-single` and still training on the other three; 1–3 seeds per cell; the zero-shot evaluation of the phase-2 checkpoints (as opposed to phase-1) is still pending; all comparisons share one actor class (flow BC + distillation), so conclusions are about the *value objective*, not about actors; the retrieval decode assumes test-time access to the offline dataset (standard in this zero-shot protocol, but worth stating). Two further qualifications. *The pretraining is not fully task-agnostic*: no reward labels are used, but the goal region is built from privileged cube coordinates, a Euclidean metric, and a hand-chosen 4 cm radius — task structure that a fully reward-free method would have to discover. *The estimator diagnosis is not fully isolated*: the density-based loss does contain a quadratic term meant to anchor its linear term, so the observed divergence implicates the estimator as implemented (finite samples, bootstrapping, an actor in the loop) rather than the linear term alone; separating those factors — and the known circularity between a learned policy and its own measure — needs controlled ablations, of which Section 7.6 supplies one (support held against loss form) and the rest remain unrun.

The strongest caveat is the last one. What this note demonstrates is a **training curriculum**, not a successor measure. The two are separable, and Section 7.8 separates them: the phase-2 object is negative on $\approx 45\%$ of the state–goal product and mis-sizes region mass by one to two orders of magnitude, while the actor reading it still scores 0.887. Success rates cannot detect this, because an actor consumes only $\arg\max_a \langle F, z\rangle$. Every claim in this note that rests on rollout success — which is every headline number — should therefore be read as a claim about the curriculum's *stability and control quality*, and none of them as evidence that the learned object integrates rewards, composes over sets, or supports the density-ratio semantics its name suggests. Section 9 argued this boundary from theory; it is now measured, and the measurement is worse than the theory predicted.

The reaching value $\mathbb{E}[\gamma^T]$ is the classical absorbing goal-conditioned value (Kaelbling, 1993) and is related to temporal-distance and contrastive goal-reaching objectives (Eysenbach et al., 2022); successor measures and their density-based TD are from Dayan (1993); Blier et al. (2021); the zero-shot protocol and the factored $F^\top B$ form are from Touati and Ollivier (2021); Touati et al. (2023); GPI from Barreto et al. (2017); the offline benchmark and goal-conditioned IQL baseline from Park et al. (2024); Kostrikov et

al. (2021); the actor from Park et al. (2025). What this note adds is deliberately narrow and empirical: on manipulation play data, (a) the standard stabilizers of successor-measure TD (orthonormality, the linear density term) are implicated in the failure rather than the fix, with the divergence reproducing on a frozen, known-good goal embedding; and (b) an absorbing-to-continuing curriculum — learn the goal-reaching value first, then re-train the forward tower into a visitation value on the frozen goal geometry — is stable where direct measure training is not. The individual components (absorbing goal values, indicator-reward policy evaluation, bilinear critics, hindsight goal regions) are all classical; the contribution, if it survives replication, is the combination and the diagnosis. It has now survived one replication: an independent re-implementation inside the FB codebase, under FB's own zero-shot protocol, reproduces the ordering and adds the control that the gain is not attributable to the privileged goal region (§7.6). A third claim must be withdrawn rather than added: the curriculum does *not* yield a usable successor measure (§7.8), and making $\tilde{M}$ non-negative by construction is a precondition for the dense-reward and set-composition programme of §10–11, not a refinement of it.

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